

Elliptic Billiards as Exact Poncelet Rotations: Monotonicity, Porisms, and the Clean-Family Obstruction

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Abstract

For the billiard inside a confocal ellipse, restricted to positively oriented orbits with an elliptic caustic, we give the exact Jacobi covering and rigid rotation number. The theorem excludes hyperbolic caustics and makes no ambient unipotent or target spectral claim.

1 The frozen confocal owner

For $0 < \varepsilon < 1$ let

$$E(\varepsilon) = \left\{ (x, y) : \varepsilon^2 x^2 + \frac{\varepsilon^2}{1 - \varepsilon^2} y^2 = 1 \right\}.$$

Its foci are $(\pm 1, 0)$. Fix $0 < f < e < 1$: $E(f)$ is the outer billiard boundary and $E(e)$ is an inner elliptic caustic. We keep the positive orientation and one-reflection clock. The corresponding Poncelet circle map is denoted $B_e^f : E(f) \rightarrow E(f)$. This is precisely the elliptic-caustic sector analyzed in the primary rotation-number sources [1, 2, 3]; hyperbolic caustics are not included.

Our convention is

$$F(\phi, e) = \int_0^\phi \frac{d\tau}{\sqrt{1 - e^2 \sin^2 \tau}}, \quad K(e) = F\left(\frac{\pi}{2}, e\right),$$

so e is the Jacobi modulus (software parameter $m = e^2$). Define

$$\omega(e, f) = \arcsin \sqrt{\frac{e^2 - f^2}{e^2(1 - f^2)}}, \tag{1}$$

$$\rho(e, f) = \frac{F(\omega(e, f), e)}{2K(e)}. \tag{2}$$

Theorem 1 (Jacobi covering and rigid rotation). *The period-one covering*

$$\pi_e^f(\theta) = \frac{1}{f} \left(-\operatorname{sn}(4K(e)\theta, e), \sqrt{1 - f^2} \operatorname{cn}(4K(e)\theta, e) \right) \tag{3}$$

satisfies

$$B_e^f \circ \pi_e^f = \pi_e^f \circ R_{\rho(e, f)}, \quad R_\rho(\theta) = \theta + \rho.$$

In particular, (2) is the rotation number.

Proof. The identity $\operatorname{sn}^2 + \operatorname{cn}^2 = 1$ puts (3) on $E(f)$, and the $4K(e)$ period gives the deck map $\theta \mapsto \theta + 1$. Set $v = 2F(\omega, e)$. The Jacobi addition formulas, together with (1), show that the line through $\pi_e^f(\theta)$ and $\pi_e^f(\theta + v/(4K(e)))$ obeys

$$C^2 = A^2/e^2 + (1 - e^2)B^2/e^2$$

when written as $Ax + By + C = 0$. This is the dual equation for tangency to $E(e)$. Positive orientation selects the next intersection, proving the conjugacy; $v/(4K(e))$ is exactly (2). This also verifies the hypotheses and conclusion of the confocal covering theorem in [1]. $\square$

References

- [1] H. E. Lomelí and J. D. Meiss, *Symmetry Reduction and Rotation Numbers for Poncelet maps*, arXiv:2309.08013v1 (2023), [doi:10.48550/arXiv.2309.08013](https://doi.org/10.48550/arXiv.2309.08013).
- [2] R. Kołodziej, *The rotation number of some transformation related to billiards in an ellipse*, *Studia Math.* 81 (1985), 293–302, [doi:10.4064/sm-81-3-293-302](https://doi.org/10.4064/sm-81-3-293-302).
- [3] S.-J. Chang and R. Friedberg, *Elliptical billiards and Poncelet's theorem*, *J. Math. Phys.* 29 (1988), 1537–1550, [doi:10.1063/1.527900](https://doi.org/10.1063/1.527900).